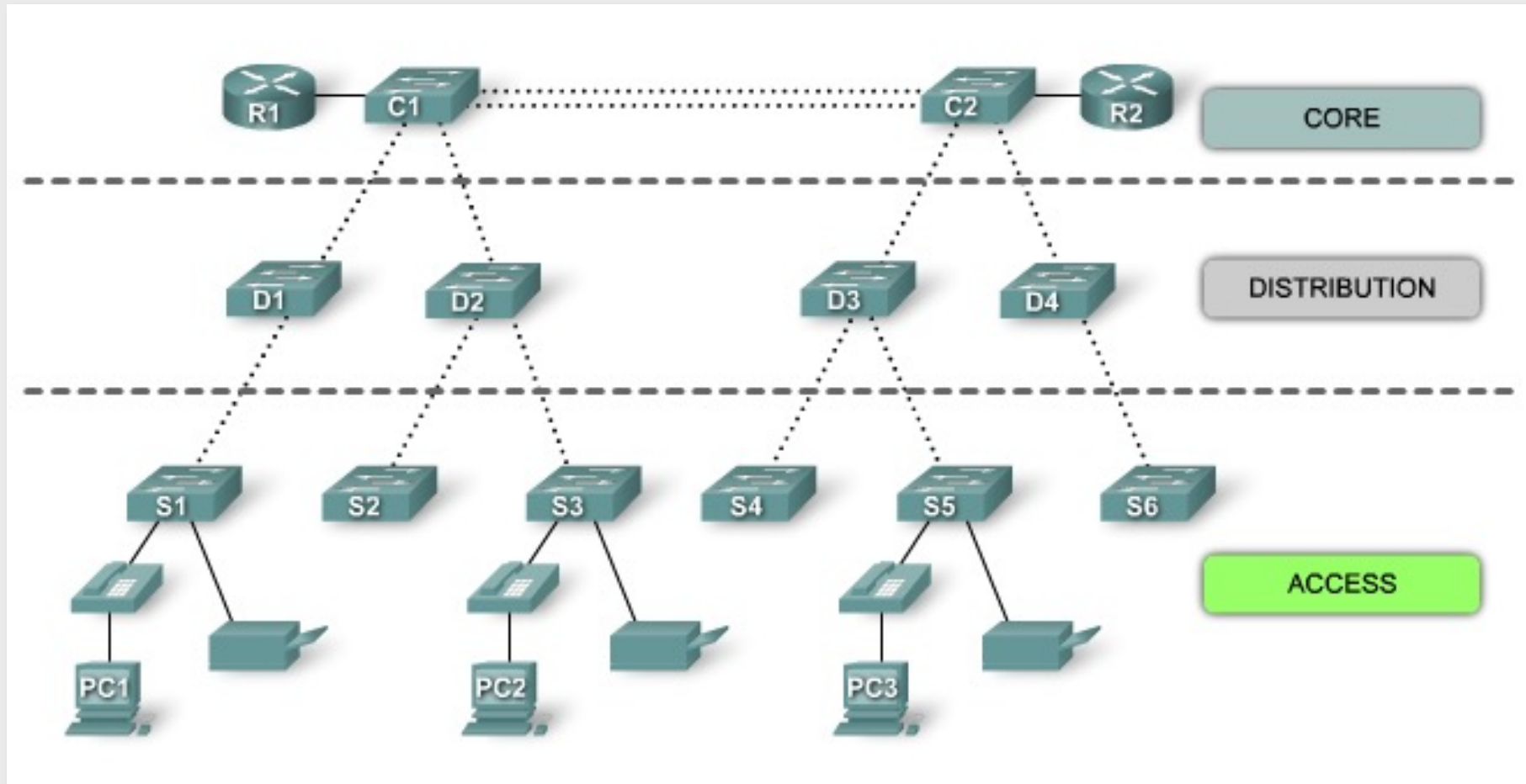


Lab 10

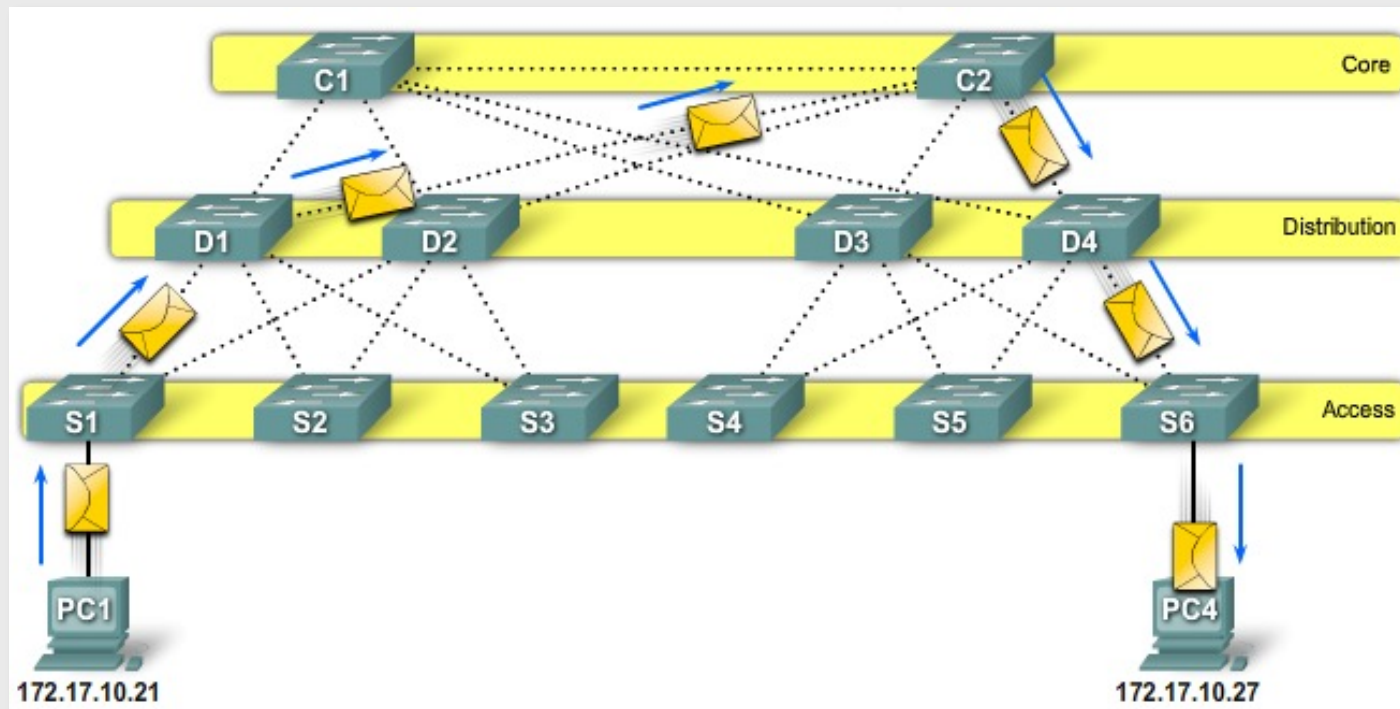
Spanning Tree Protocol

LAN hierarchical design



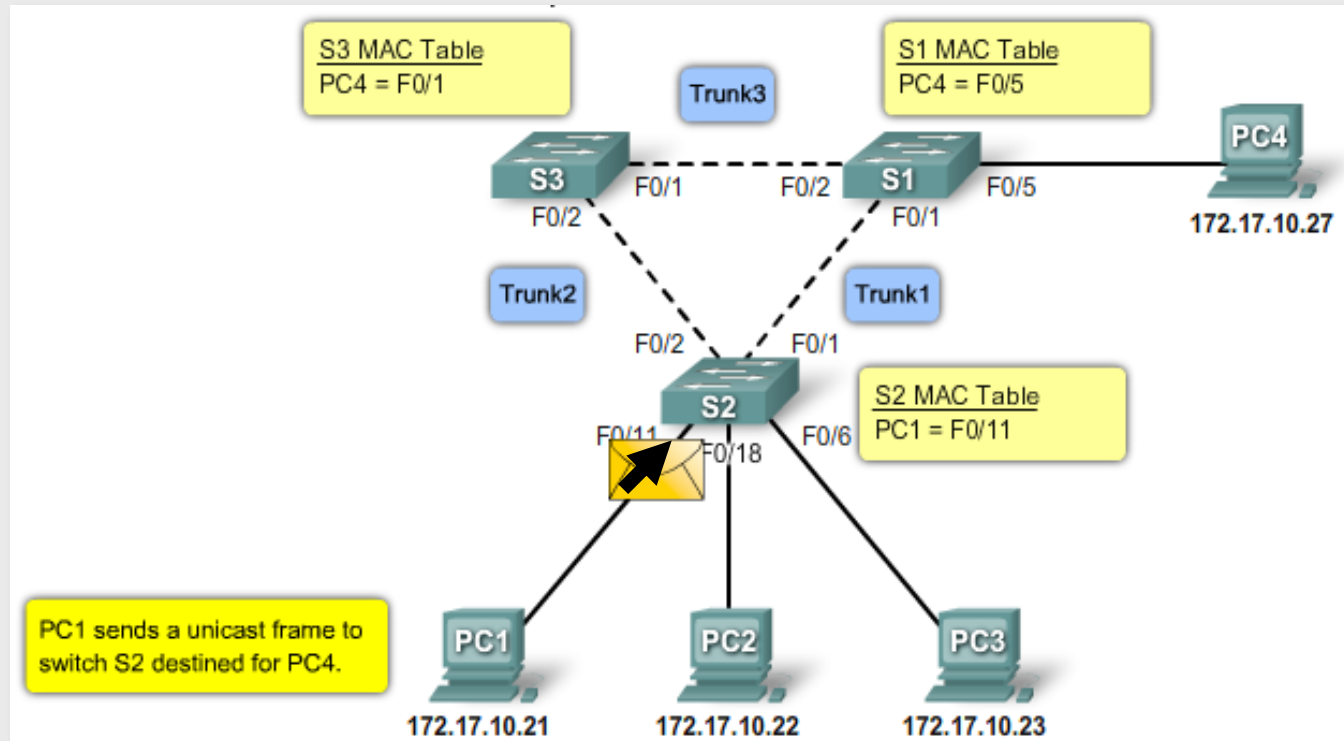
Redundant design

- Additional hardware plus multiple paths
- Accommodate a single point of failure
 - Path failure (access to distribution or distribution to core)
 - Switch failure (distribution or core)



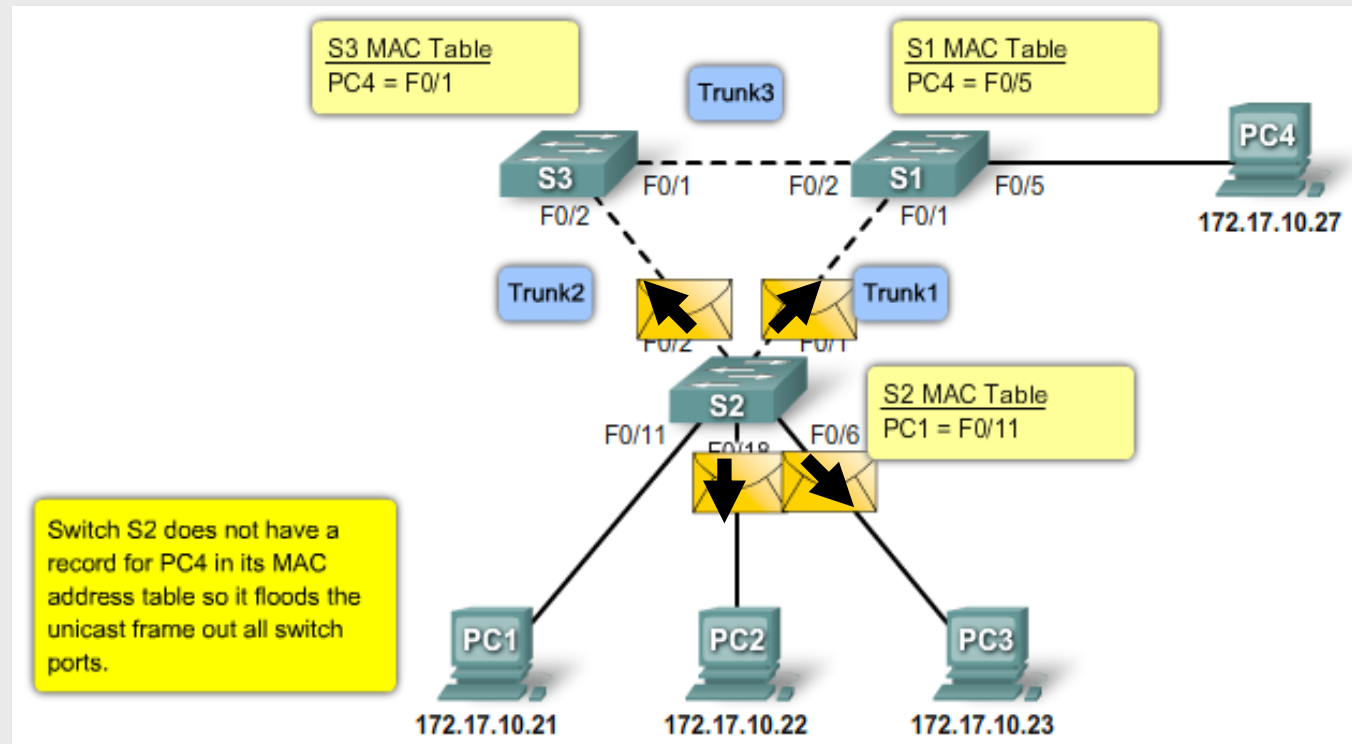
Issues with redundancy

- Duplicate unicast frames



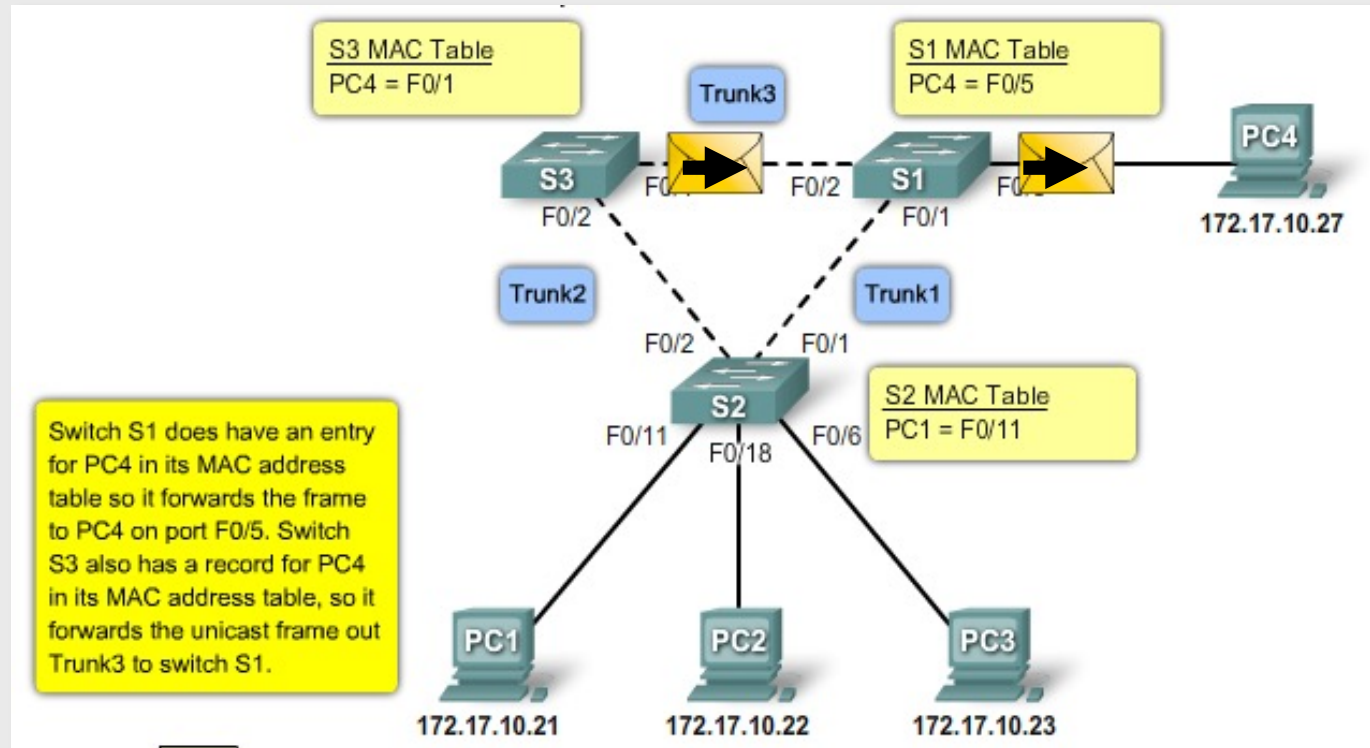
Issues with redundancy

- Duplicate unicast frames



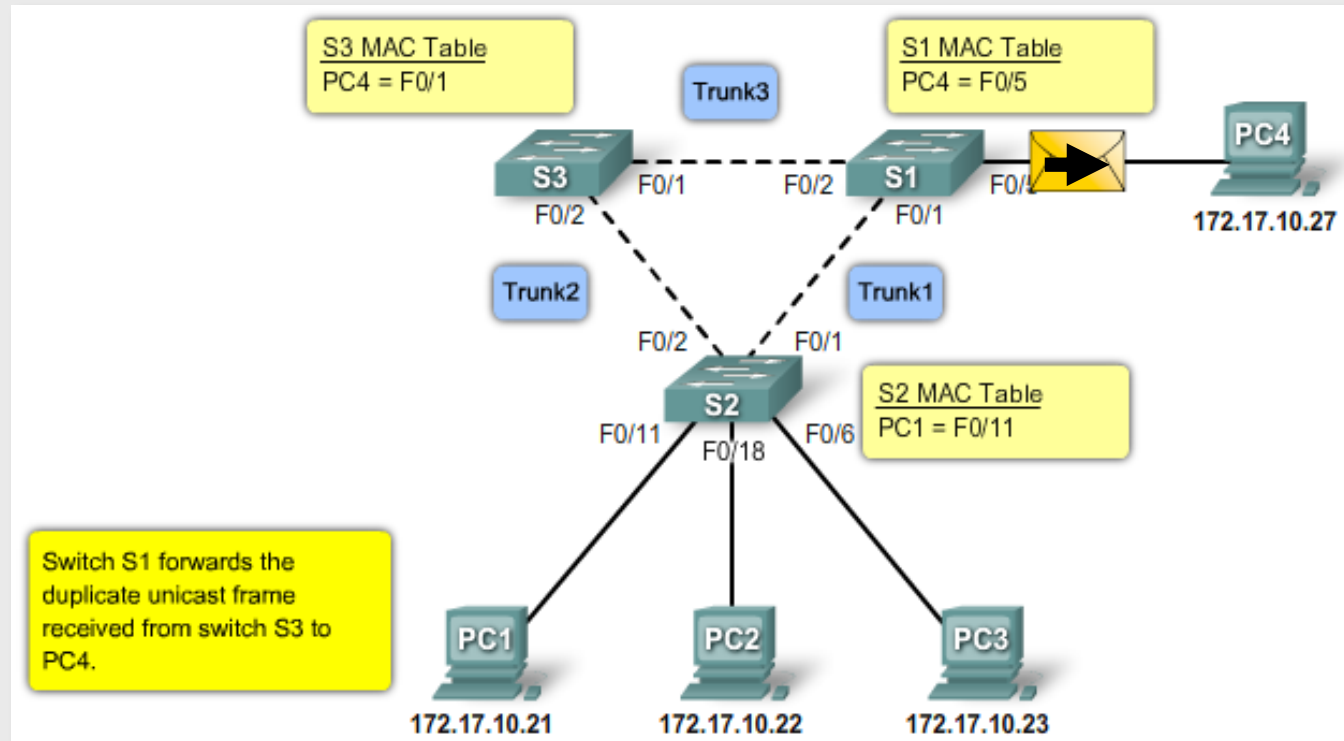
Issues with redundancy

- Duplicate unicast frames



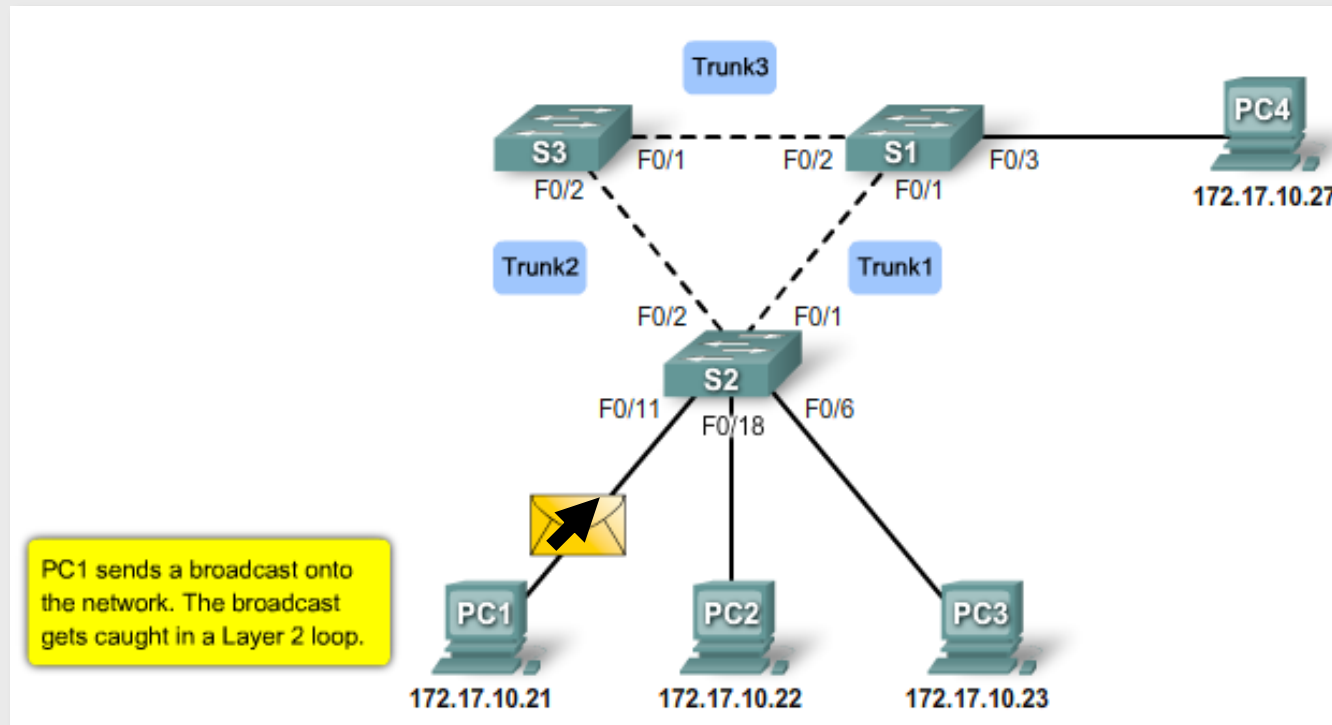
Issues with redundancy

- Duplicate unicast frames



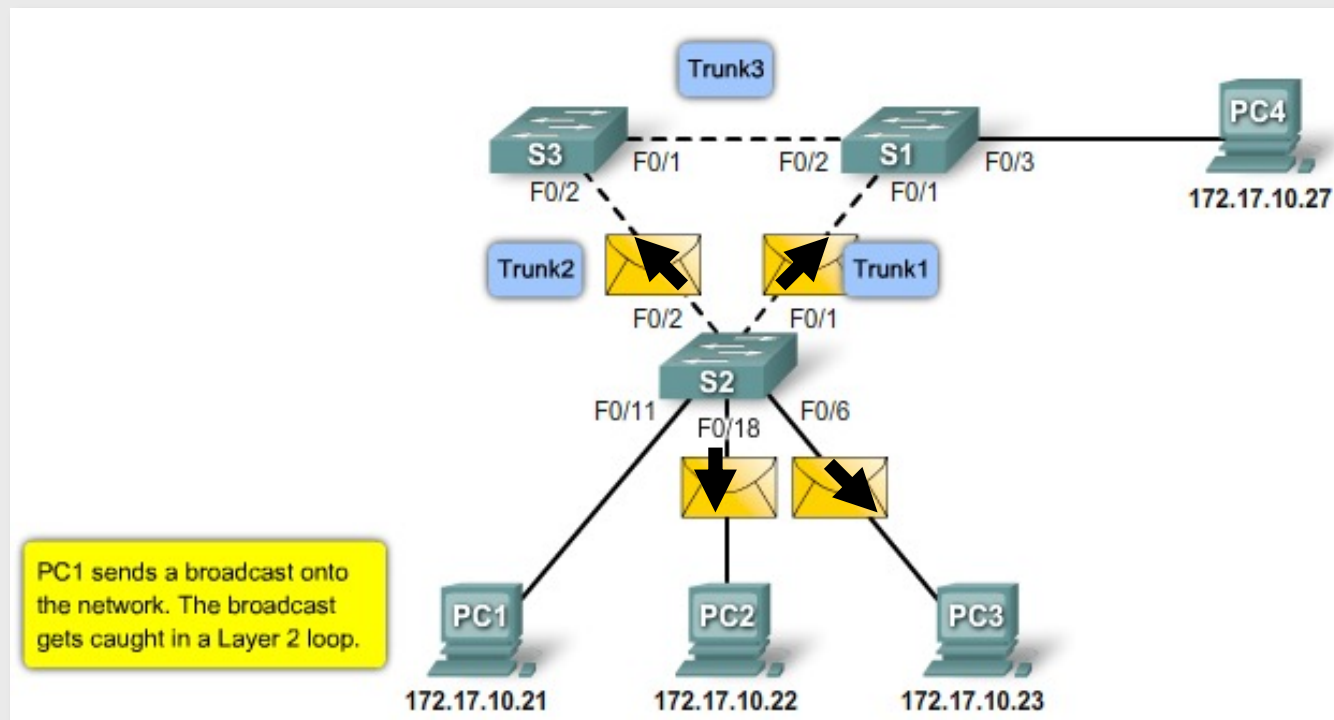
Issues with redundancy

- Broadcast storms



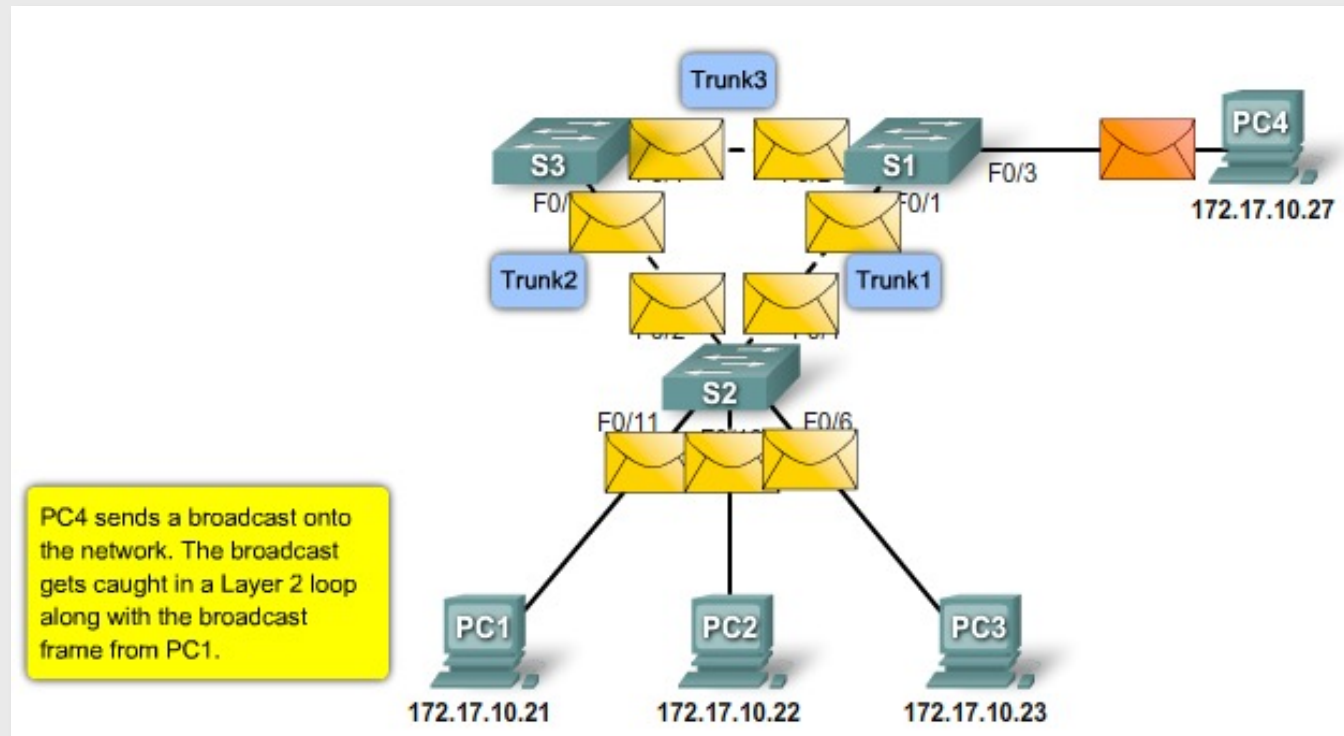
Issues with redundancy

- Broadcast storms



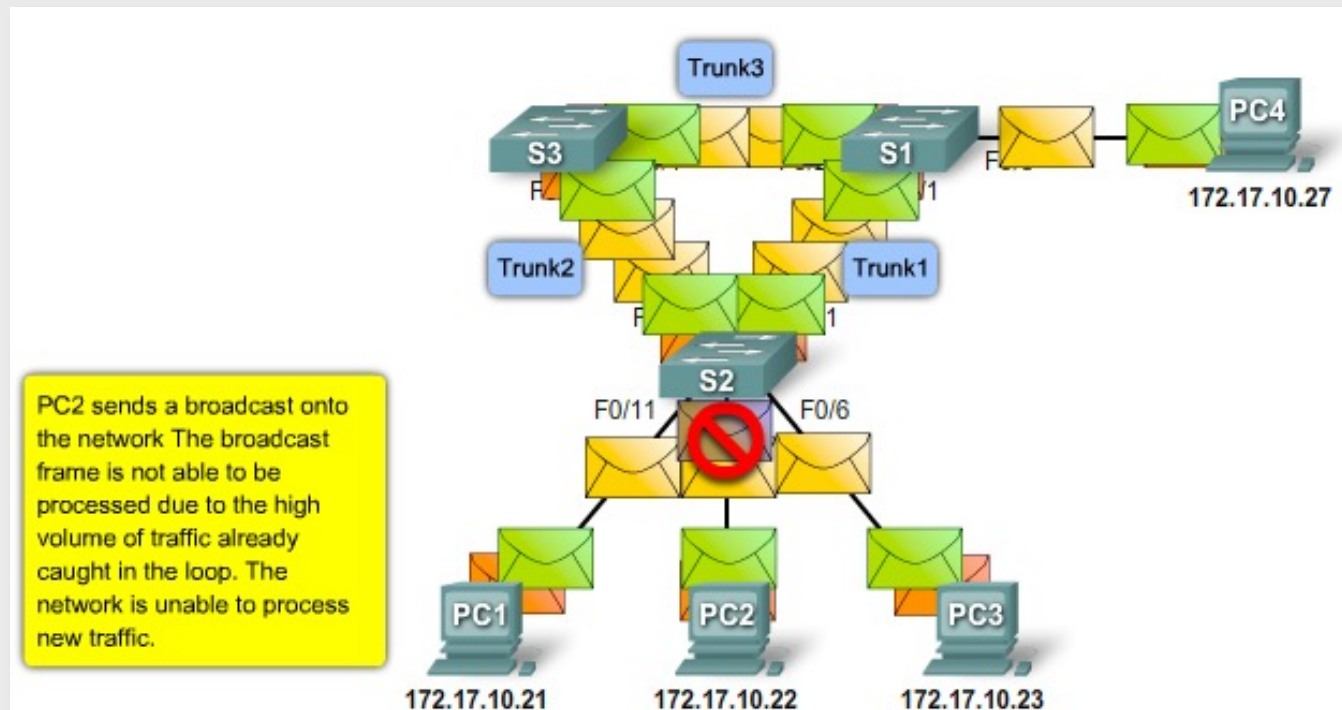
Issues with redundancy

- Broadcast storms



Issues with redundancy

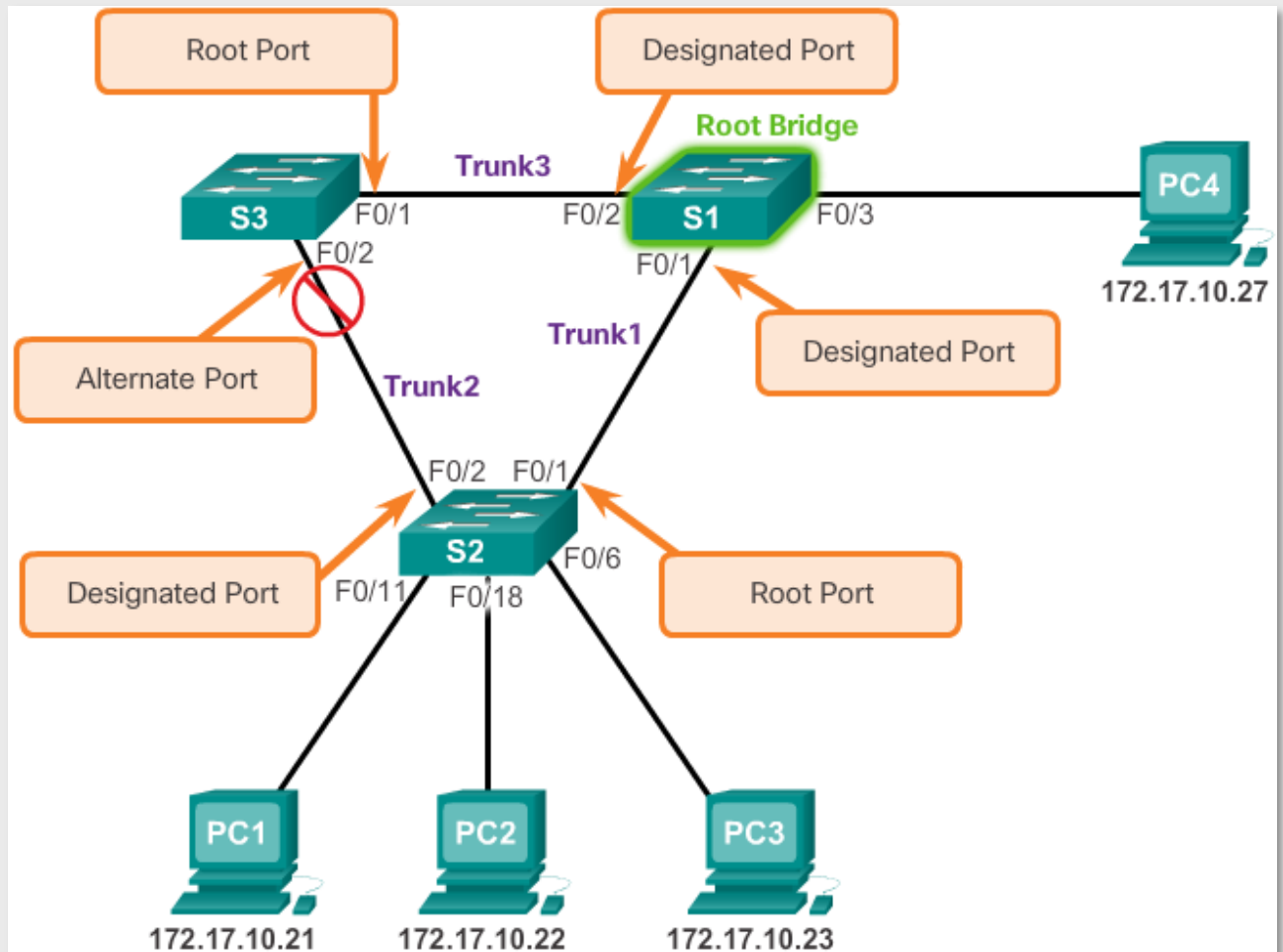
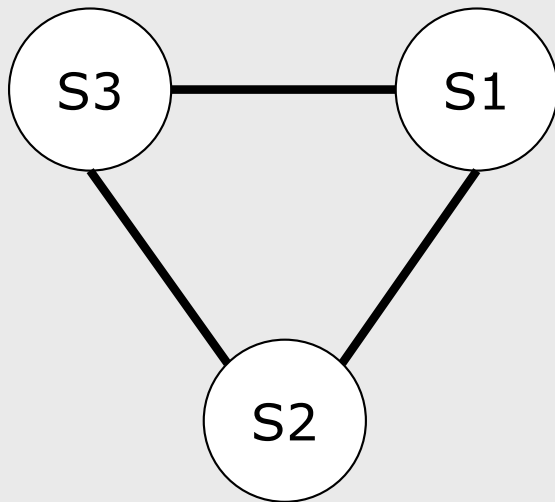
- Broadcast storms



The Spanning Tree Protocol (STP) addresses loop issues, ensuring that **there is only one logical path between all destinations on the network** by intentionally blocking redundant paths that could cause a loop

Spanning Tree Protocol

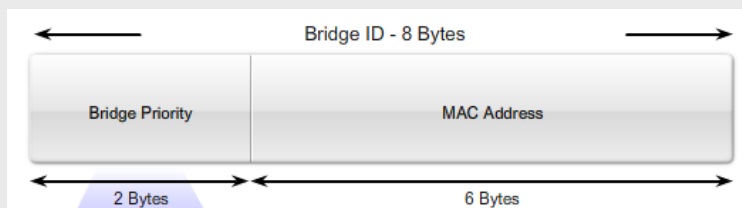
- Basic concept: block frame forwarding over selected ports to avoid loops
 - Fully distributed protocol, goes through three phases
 - Root bridge election
 - Root port selection
 - Designated port selection
-
- ```
graph LR; S2[S2] ---|Trunk3| S1[S1]; S2 -- Root Port --> E0_1[E0/1]; S1 -- Designated Port --> E0_2[E0/2]; S2 --> E0_3_2[E0/3]; S1 --> E0_3_1[E0/3]; style S1 stroke:#00FF00,stroke-width:2px; linkStyle 0 stroke:#FFA500,stroke-width:2px;
```



# Bridge Protocol Data Unit (BPDU)

Configuration BPDU

| Field # | Bytes | Field         |
|---------|-------|---------------|
| 1-4     | 2     | Protocol ID   |
|         | 1     | Version       |
|         | 1     | Message type  |
|         | 1     | Flags         |
| 5-8     | 8     | Root ID       |
|         | 4     | Cost of path  |
|         | 8     | Bridge ID     |
|         | 2     | Port ID       |
| 9-12    | 2     | Message age   |
|         | 2     | Max age       |
|         | 2     | Hello time    |
|         | 2     | Forward delay |



## ■ Flags

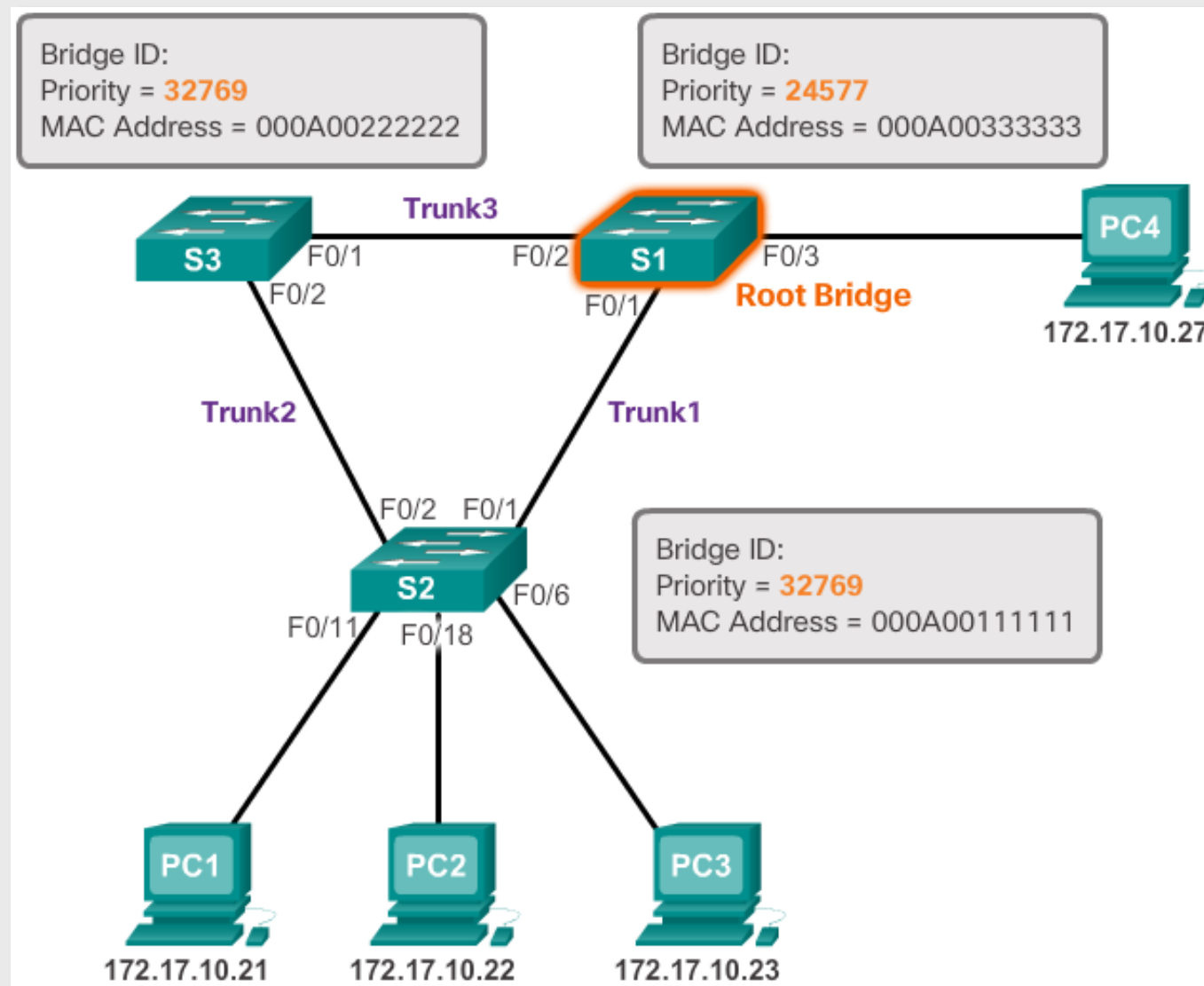
- TC: Topology Change
- TCA: Topology Change Acknowledgement
- **Root ID**: the identifier of the (candidate) root bridge
- **Bridge ID**: the identifier of the bridge transmitting the BPDU
- **Root path cost**: cost of the path to the root bridge from the bridge sending the BPDU
- **Port ID**: the identifier of the port through which the BPDU is sent by the bridge
- **Message age**: time elapsed since the root sent the first configuration BPDU
- **Max age**: max time of validity, after which the STP process must be restarted
- **Hello time**: time period between two root bridge configuration BPDUs
- **Forward delay**: controls port state transitions

# Root bridge election

---

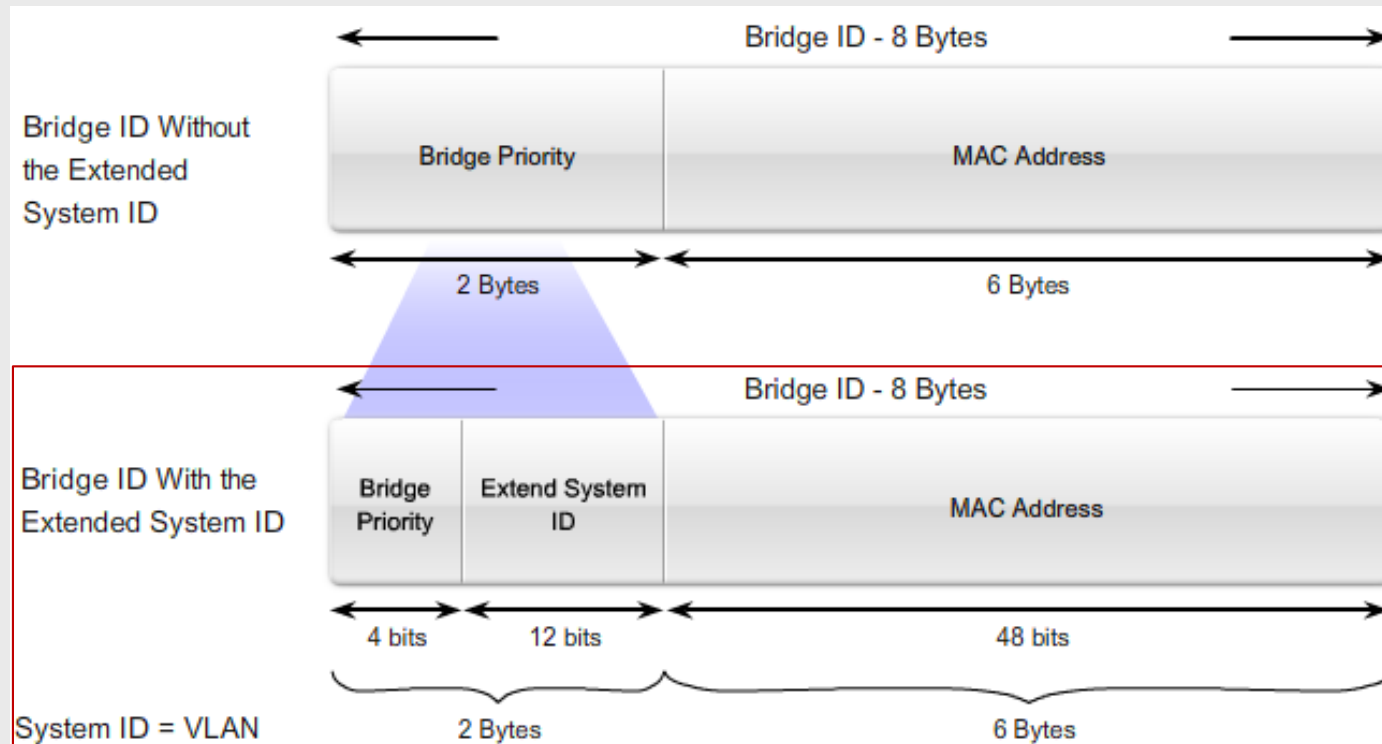
- Each switch assumes to be the root bridge, and advertises its Bridge ID as the Root ID in BPDUs sent periodically (2s default) over all ports
- If the switch receives a BPDU with a **lower** Root ID
  - the root ID field is updated on the switch
  - the Root path cost is updated by adding the Path cost associated to the ingress port (more on this later on)
  - the new root ID and path cost are advertised in all future BPDU frame transmissions
- After some time, there will be only one switch generating configuration BPDUs with Root ID equal to the switch Bridge ID
  - the root is elected

# Root bridge election – example



# Bridge ID

## ■ Bridge ID definition



Multiple STP instances  
(one per VLAN)

This format is Cisco  
specific

## ■ Bridge priority default value: 32768

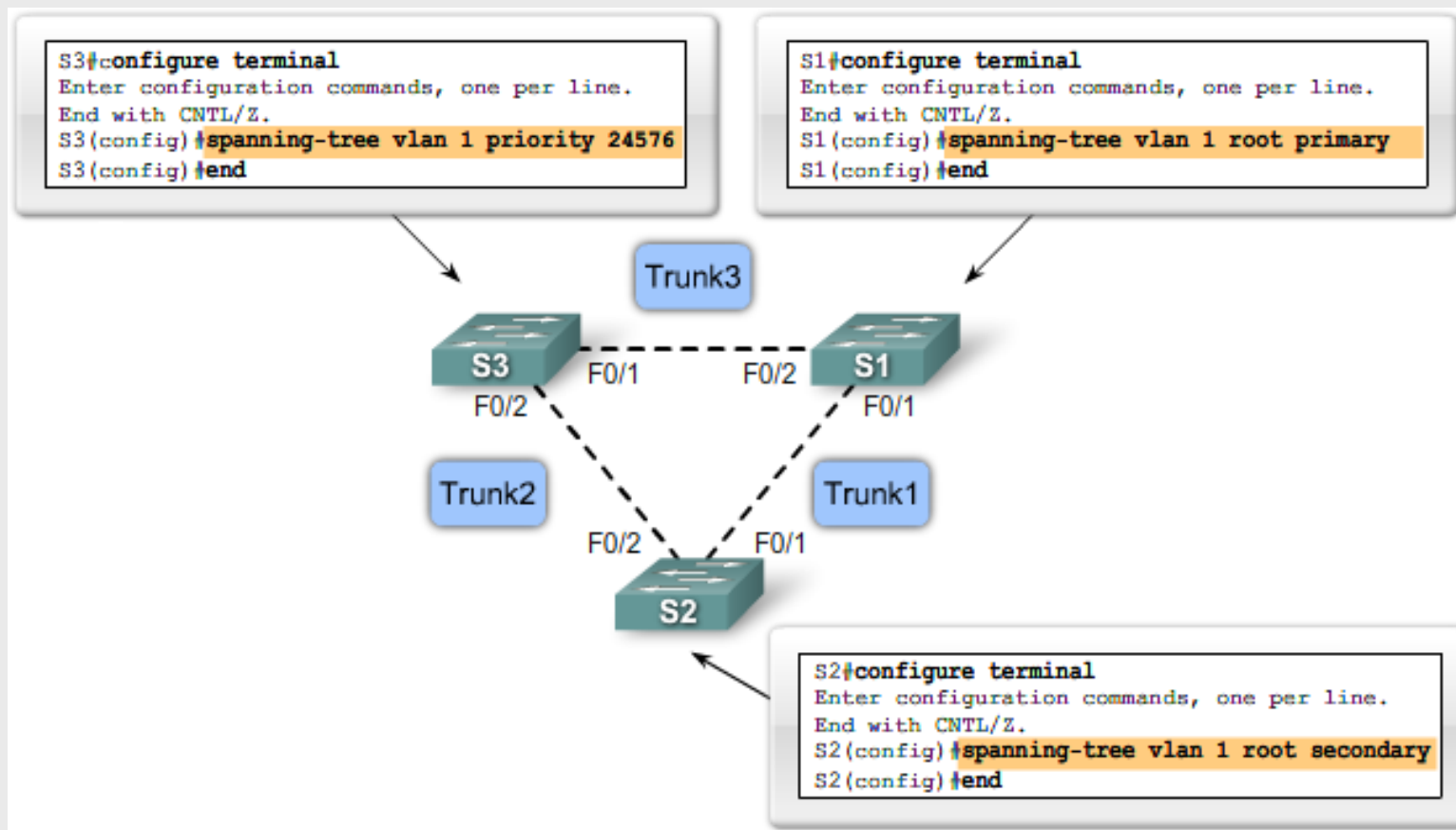
- Increment or decrement by multiple values of 4096 (recommended practice)



# Bridge ID

## ■ Configure the priority

```
Sw1(config)#spanning-tree vlan vlan-id priority value
Sw1(config)#spanning-tree vlan vlan-id root primary
Sw1(config)#spanning-tree vlan vlan-id root secondary
```



# Bridge ID

## ■ Verify configuration

```
S1#show spanning-tree
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID Priority 24577
 Address 000A.0033.3333
```

```
This bridge is the root
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID Priority 24577 (priority 24576 sys-id-ext 1)
```

```
Address 000A.0033.3333
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Aging Time 300
```

```
Interface Role Sts Cost Prio.Nbr Type
```

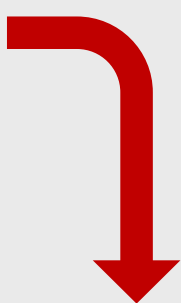
```

Fa0/1 Desg FWD 4 128.1 P2p
```

```
Fa0/2 Desg FWD 4 128.2 P2p
```

```
S1#
```

# Root port selection

- The root port is the switch port with the **lowest path cost** to the root bridge
    - Every switch has a single root port defined, except for the root bridge
  - Root path cost associated to a port = Root path cost included in the received BPDU + path cost associated to the port
  - Tie break rules
    - Received BPDU includes the lowest Bridge ID
    - Received BPDU includes the lowest Port ID
  - This step is executed concurrently with root bridge election!
    - Path costs are updated during BPDU exchange
    - Root port is determined accordingly
    - Root port may change multiple times before STP converges
- 

# Path cost

- Default port costs are defined by the port speed

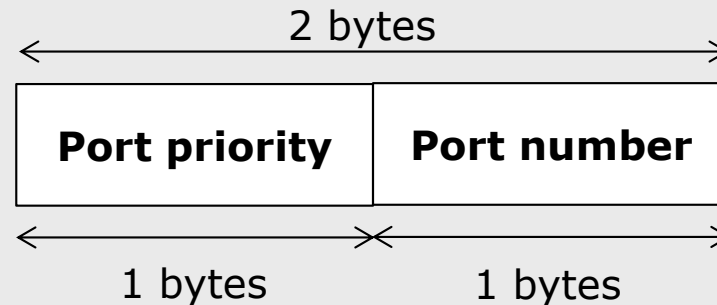
| Link Speed | Cost (Revised IEEE Specification) |
|------------|-----------------------------------|
| 10 Gb/s    | 2                                 |
| 1 Gb/s     | 4                                 |
| 100 Mb/s   | 19                                |
| 10 Mb/s    | 100                               |

- Port cost may be reconfigured to control the spanning tree formation

```
S2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#interface f0/1
S2(config-if)#spanning-tree cost 25
S2(config-if)#end
S2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#interface f0/1
S2(config-if)#no spanning-tree cost
S2(config-if)#end
S2#
```

# Port ID

- Port ID definition

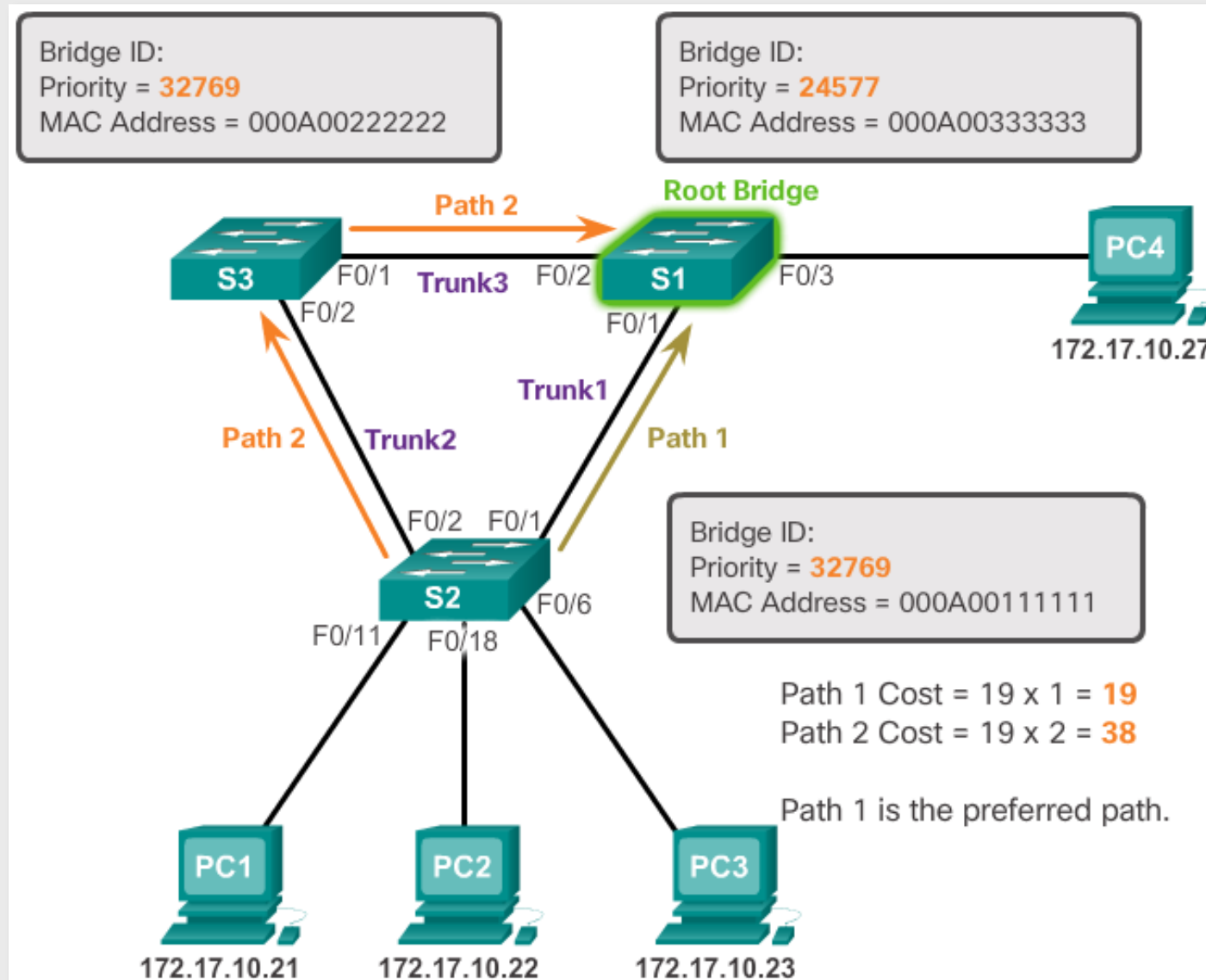


- Port priority default value: 128
  - Increment or decrement by multiple values of 16 (recommended)

- Configure priority

```
S2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#interface f0/1
S2(config-if)#spanning-tree port-priority 112
S2(config-if)#end
S2#
```

# Root port selection



# Root port selection

```
S3#show spanning-tree
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID Priority 24577
Address 000A.0033.3333
Cost 19
Port 1(FastEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 000A.0022.2222
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
```

| Interface | Role | Sts | Cost | Prio.Nbr | Type |
|-----------|------|-----|------|----------|------|
| Fa0/1     | Root | FWD | 19   | 128.1    | P2p  |
| Fa0/2     | Altn | BLK | 19   | 128.2    | P2p  |

```
S1#show spanning-tree
```

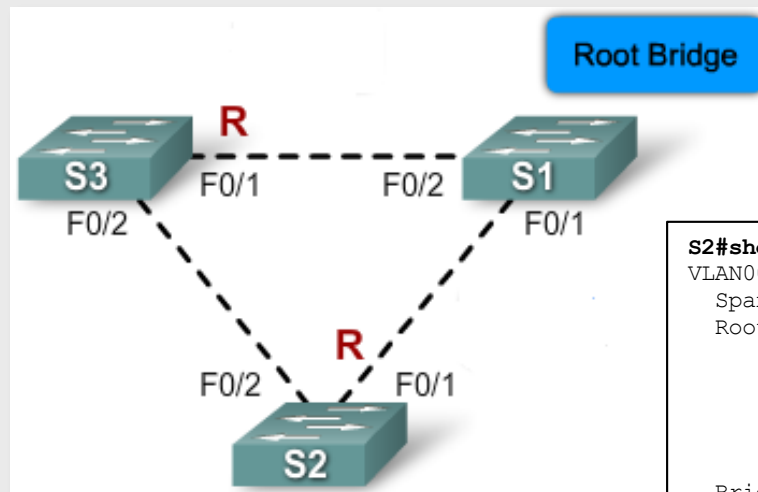
```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID Priority 24577
Address 000A.0033.3333
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID Priority 24577 (priority 24576 sys-id-ext 1)
Address 000A.0033.3333
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
```

| Interface | Role | Sts | Cost | Prio.Nbr | Type |
|-----------|------|-----|------|----------|------|
| Fa0/2     | Desg | FWD | 19   | 128.2    | P2p  |
| Fa0/1     | Desg | FWD | 19   | 128.1    | P2p  |



```
S2#show spanning-tree
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID Priority 24577
Address 000A.0033.3333
Cost 19
Port 1(FastEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 000A.0011.1111
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
```

| Interface | Role | Sts | Cost | Prio.Nbr | Type |
|-----------|------|-----|------|----------|------|
| Fa0/1     | Root | FWD | 19   | 128.1    | P2p  |
| Fa0/2     | Desg | FWD | 19   | 128.2    | P2p  |

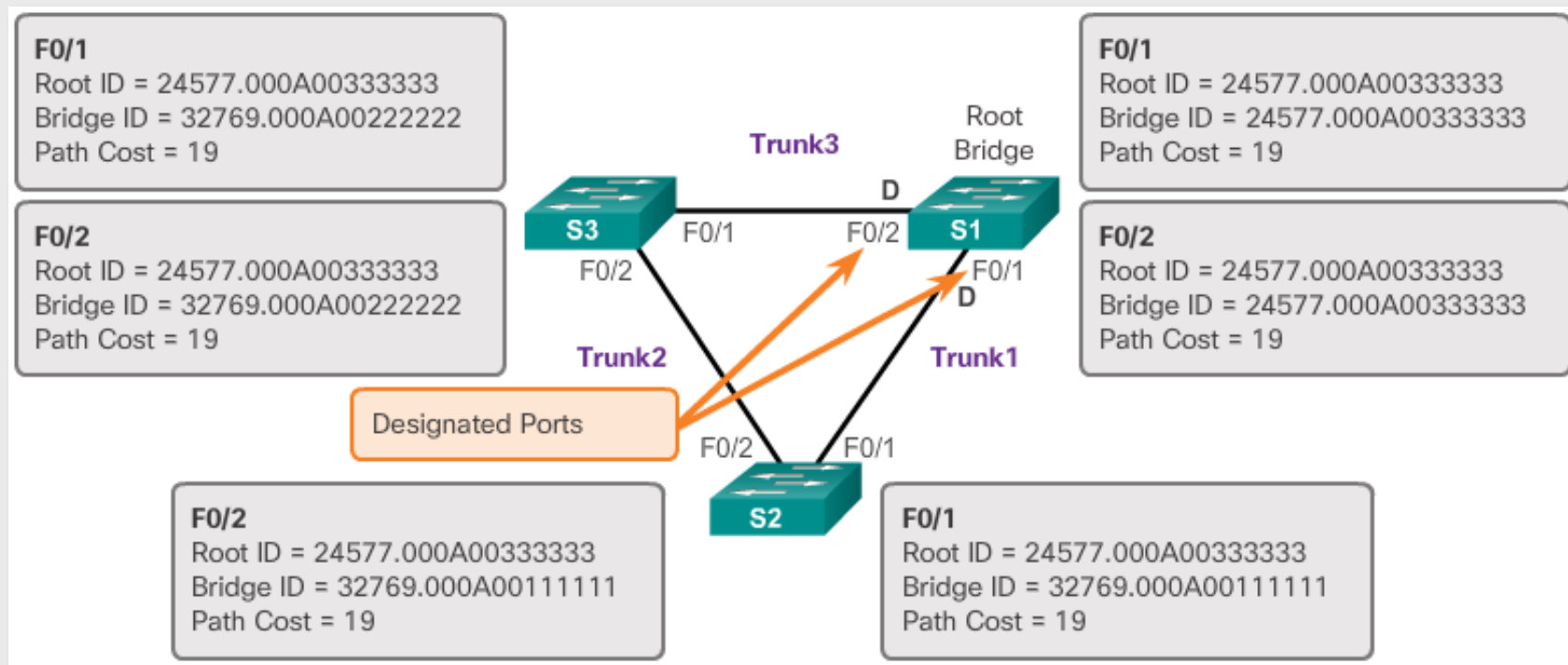
# Designated port selection

- The root bridge configures all of its ports as designated
- If a switch does not receive BPDUs on a given non-root port, the port is configured as designated (i.e., forwarding)
- If a switch receives BPDUs on a given non-root port, a competition for port designation must occur
  - Only one non-root port shall be designated, i.e., forwarding, on a given LAN segment, otherwise a loop occurs
- Competition is won by the switch port through which the BPDU with the highest priority is transmitted
  - **Lowest** tuple <root path cost, bridge ID, Port ID>
- The winning switch port is configured as designated
- All other ports are configured as alternate ports, i.e., in a blocking state



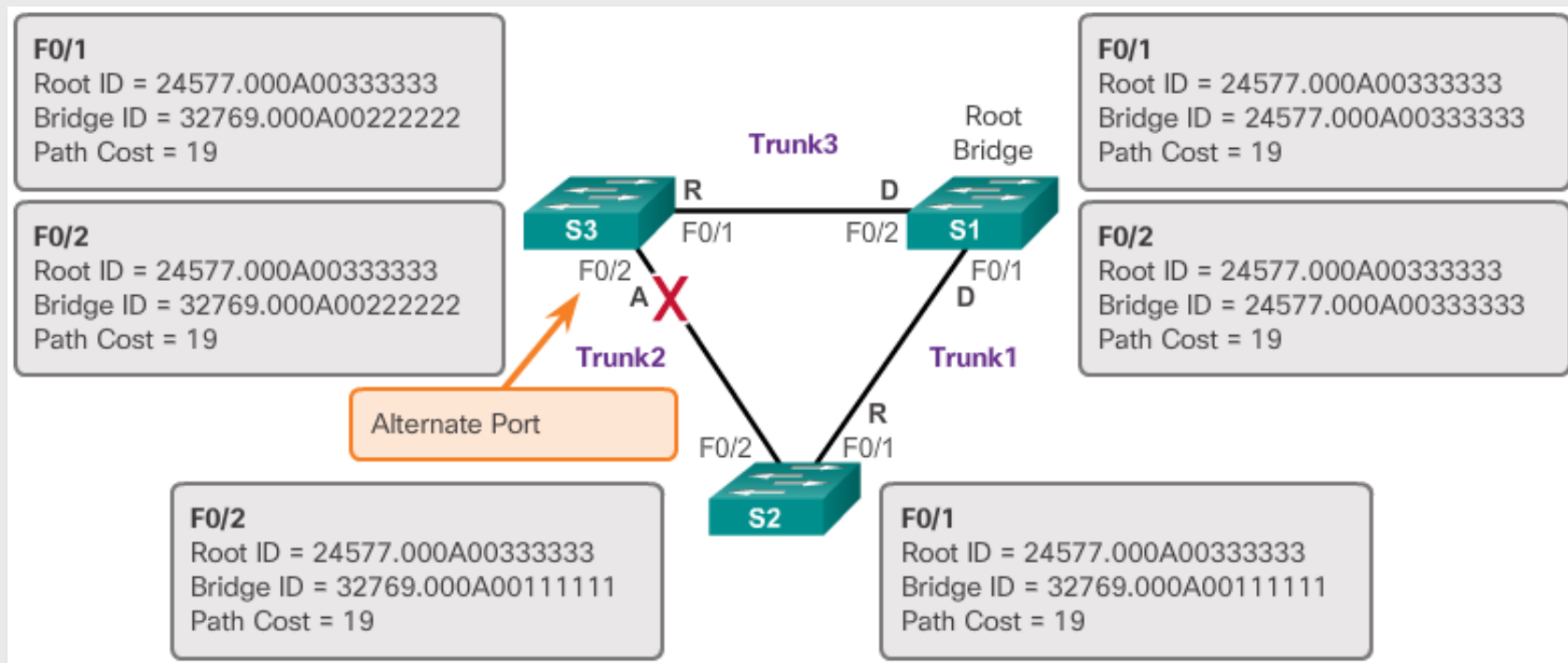
# Designated port selection

- S1 (root bridge) configures both of its trunk ports as designated ports



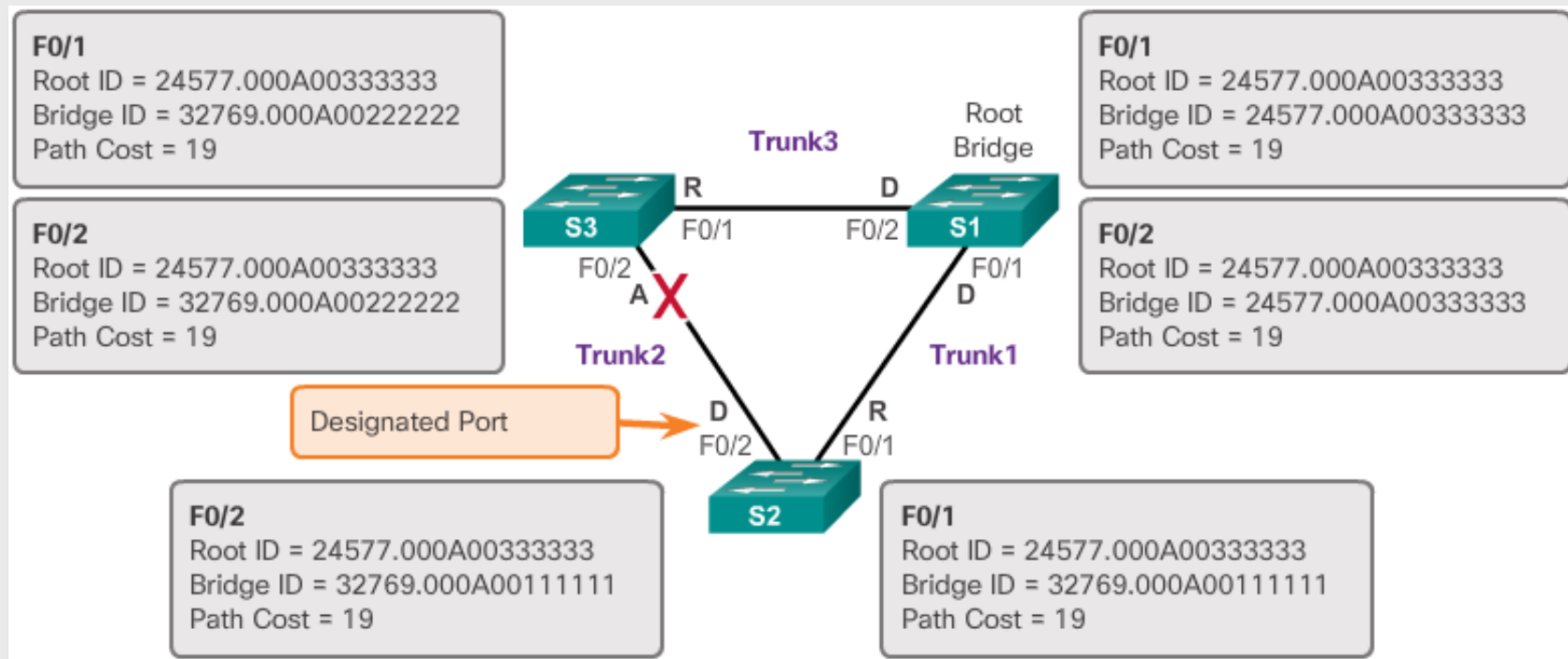
# Designated port selection

- S2 and S3 exchange BPDUs. S3 identifies S2 as having the same root path cost but a lower Bridge ID. S3 configures port F0/2 as an alternate port.



# Designated port selection

- S2 configures port F0/2 as a designated port.



# Designated port selection

## S3#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID      Priority      24577  
 Address      000A.0033.3333  
 Cost          19  
 Port          1(FastEthernet0/1)  
 Hello Time    2 sec    Max Age 20 sec    Forward Delay 15 sec

Bridge ID    Priority      32769 (priority 32768 sys-id-ext 1)  
 Address      000A.0022.2222  
 Hello Time    2 sec    Max Age 20 sec    Forward Delay 15 sec  
 Aging Time    20

| Interface    | Role        | Sts        | Cost      | Prio.Nbr     | Type       |
|--------------|-------------|------------|-----------|--------------|------------|
| Fa0/1        | Root        | FWD        | 19        | 128.1        | P2p        |
| <b>Fa0/2</b> | <b>Altn</b> | <b>BLK</b> | <b>19</b> | <b>128.2</b> | <b>P2p</b> |

## S1#show spanning-tree

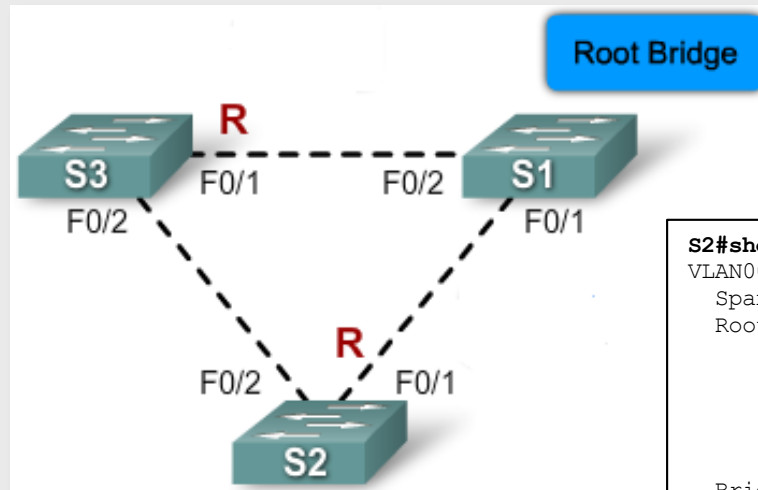
VLAN0001

Spanning tree enabled protocol ieee

Root ID      Priority      24577  
 Address      000A.0033.3333  
 This bridge is the root  
 Hello Time    2 sec    Max Age 20 sec    Forward Delay 15 sec

Bridge ID    Priority      24577 (priority 24576 sys-id-ext 1)  
 Address      000A.0033.3333  
 Hello Time    2 sec    Max Age 20 sec    Forward Delay 15 sec  
 Aging Time    20

| Interface    | Role        | Sts        | Cost      | Prio.Nbr     | Type       |
|--------------|-------------|------------|-----------|--------------|------------|
| <b>Fa0/2</b> | <b>Desg</b> | <b>FWD</b> | <b>19</b> | <b>128.2</b> | <b>P2p</b> |
| <b>Fa0/1</b> | <b>Desg</b> | <b>FWD</b> | <b>19</b> | <b>128.1</b> | <b>P2p</b> |



## S2#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID      Priority      24577  
 Address      000A.0033.3333  
 Cost          19  
 Port          1(FastEthernet0/1)  
 Hello Time    2 sec    Max Age 20 sec    Forward Delay 15 sec

Bridge ID    Priority      32769 (priority 32768 sys-id-ext 1)  
 Address      000A.0011.1111  
 Hello Time    2 sec    Max Age 20 sec    Forward Delay 15 sec  
 Aging Time    20

| Interface    | Role        | Sts        | Cost      | Prio.Nbr     | Type       |
|--------------|-------------|------------|-----------|--------------|------------|
| Fa0/1        | Root        | FWD        | 19        | 128.1        | P2p        |
| <b>Fa0/2</b> | <b>Desg</b> | <b>FWD</b> | <b>19</b> | <b>128.2</b> | <b>P2p</b> |

# RSTP and other STP variants

## ■ Rapid Spanning Tree Protocol

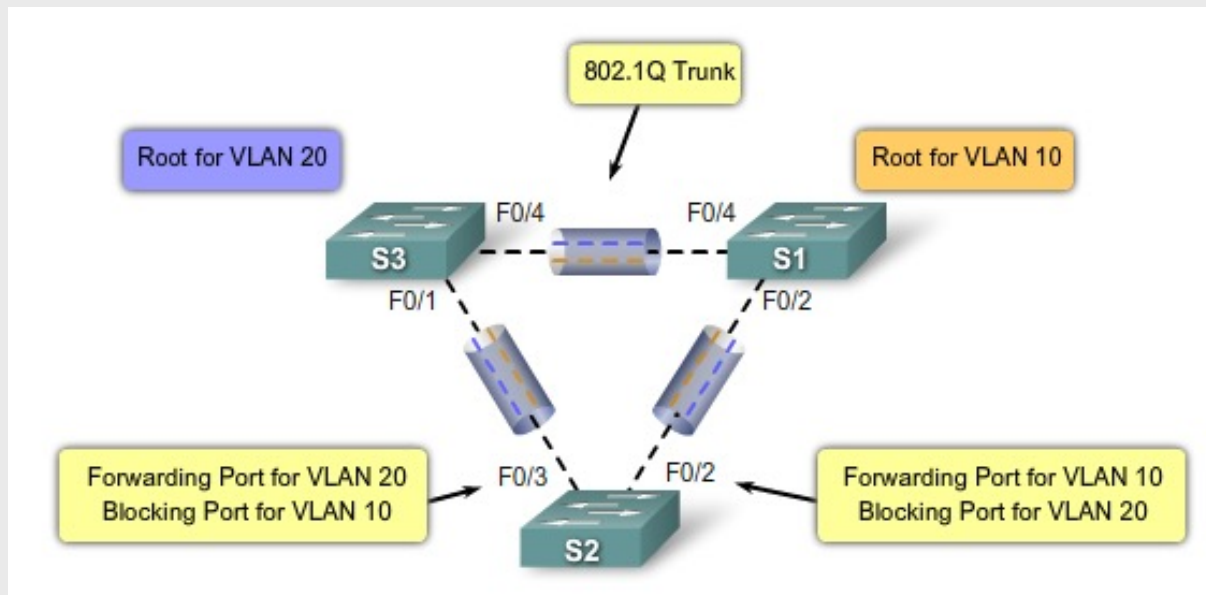
- Faster convergence: respond to changes within  $3 \times$  Hello times (default: 3 times 2 seconds) or within a few hundred milliseconds of a physical link failure
- Requires additional assumptions
  - All switches run RSTP
  - Only p2p connections between switches
- Originally std IEEE 802.1w, is now incorporated into std IEEE 802.1D-2004 and obsoletes the original STP!

## ■ Multiple STP

- "Per-VLAN" Multiple Spanning Tree Protocol
  - configures a separate Spanning Tree for each VLAN group
- Originally std IEEE 802.1s, now incorporated into std IEEE 802.1Q-2005

# RSTP and other STP variants

- Cisco proprietary STP variants
  - Per-VLAN STP (PVST), PVST+, Rapid-PVST+
- Per-VLAN spanning tree protocol plus (PVST+)
  - Provides support for IEEE 802.1Q trunking
  - Dedicated STP instance per VLAN
    - Allows for fine tuning of bandwidth resources

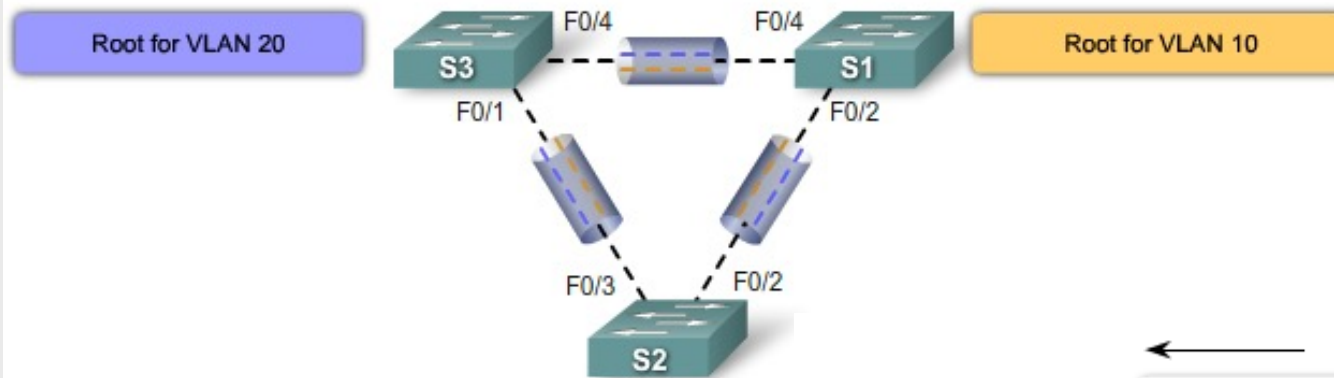


# PVST+ Bridge ID

Priority + VLAN ID + MAC Address = BID

32768 + 10 + 000A00333333 = 32778.000A00333333

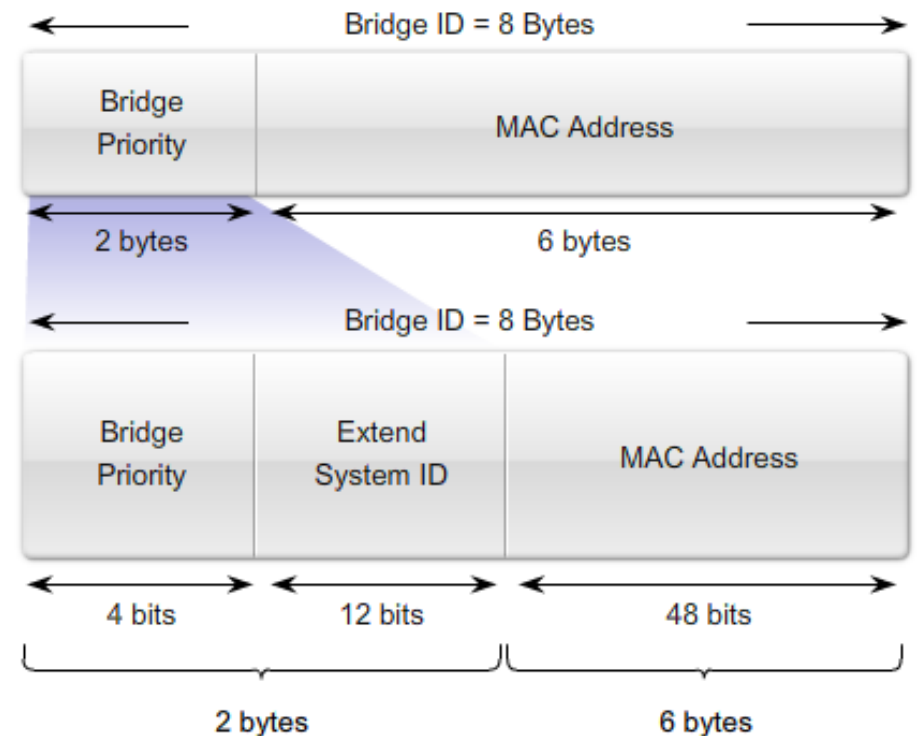
32768 + 20 + 000A00333333 = 32788.000A00333333



Bridge ID without the extended system ID

Extended bridge ID with system ID

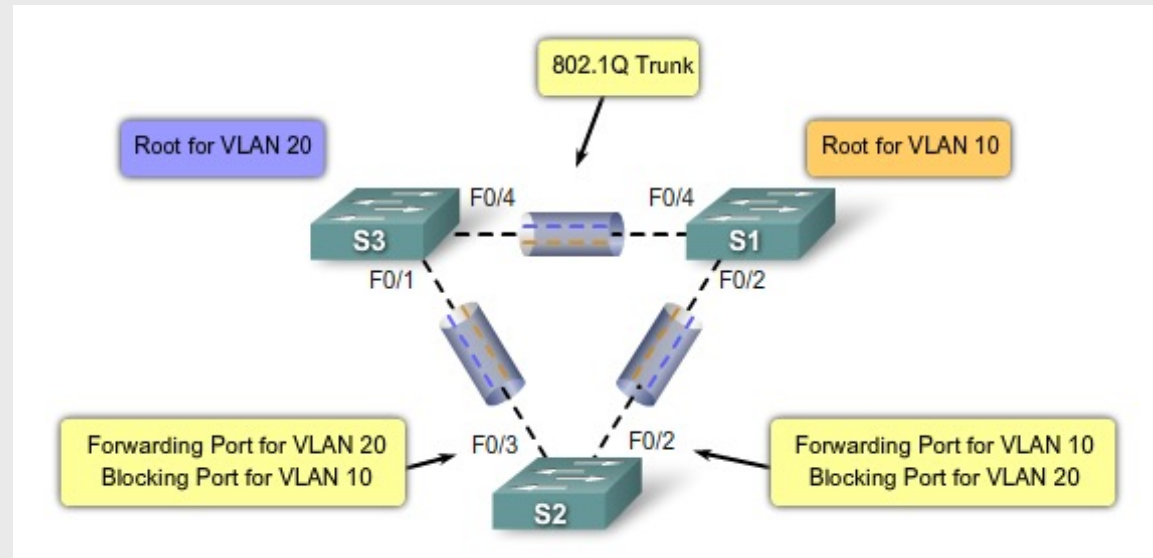
System ID = VLAN



# PVST+ configuration

```
S3(config)#spanning-tree vlan 20 root primary
S3(config)#spanning-tree vlan 10 root secondary

S1(config)#spanning-tree vlan 10 root primary
S1(config)#spanning-tree vlan 20 root secondary
```



```
S1#show spanning-tree active
<output omitted>
VLAN0010
 Spanning tree enabled protocol ieee
 Root ID Priority 4106
 Address 0019.aa9e.b000
 This bridge is the root
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Bridge ID Priority 4106 (priority 4096 sys-id-ext 10)
 Address 0019.aa9e.b000
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300
 Interface Role Sts Cost Prio.Nbr Type

 Fa0/2 Desg FWD 19 128.2 P2p
 Fa0/4 Desg FWD 19 128.4 P2p
<output omitted>
```

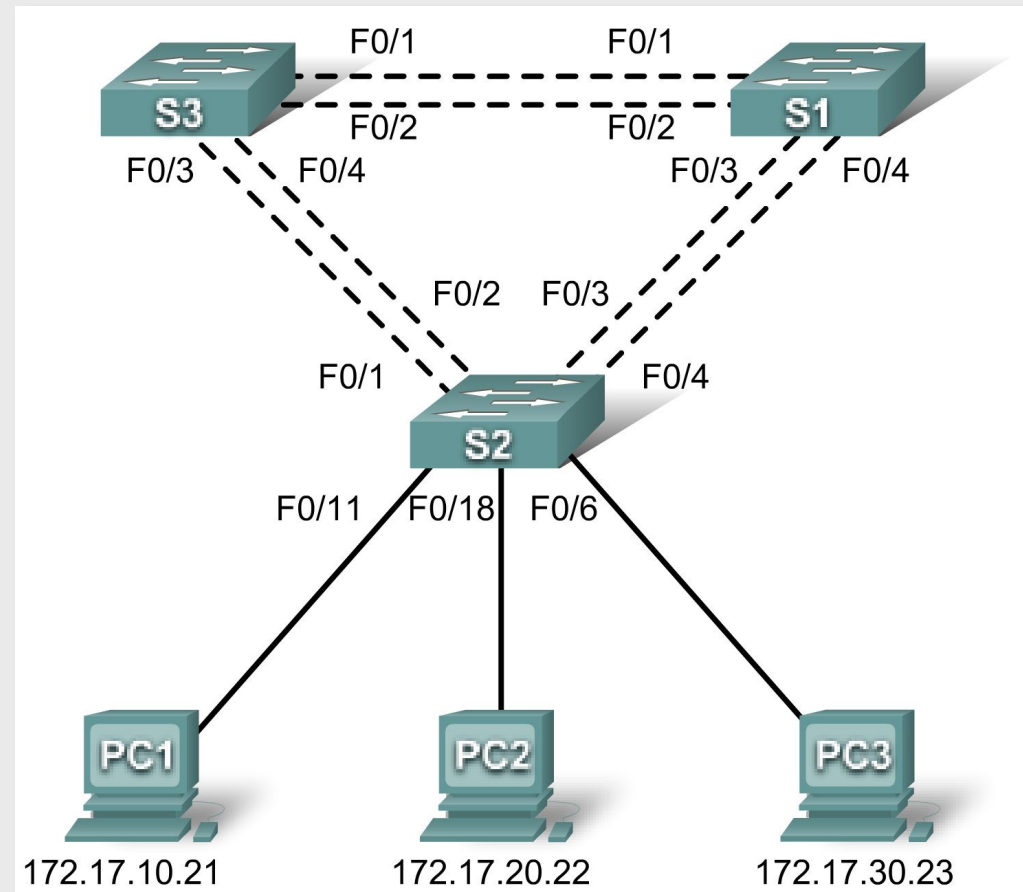


# Catalyst 2960 default configuration

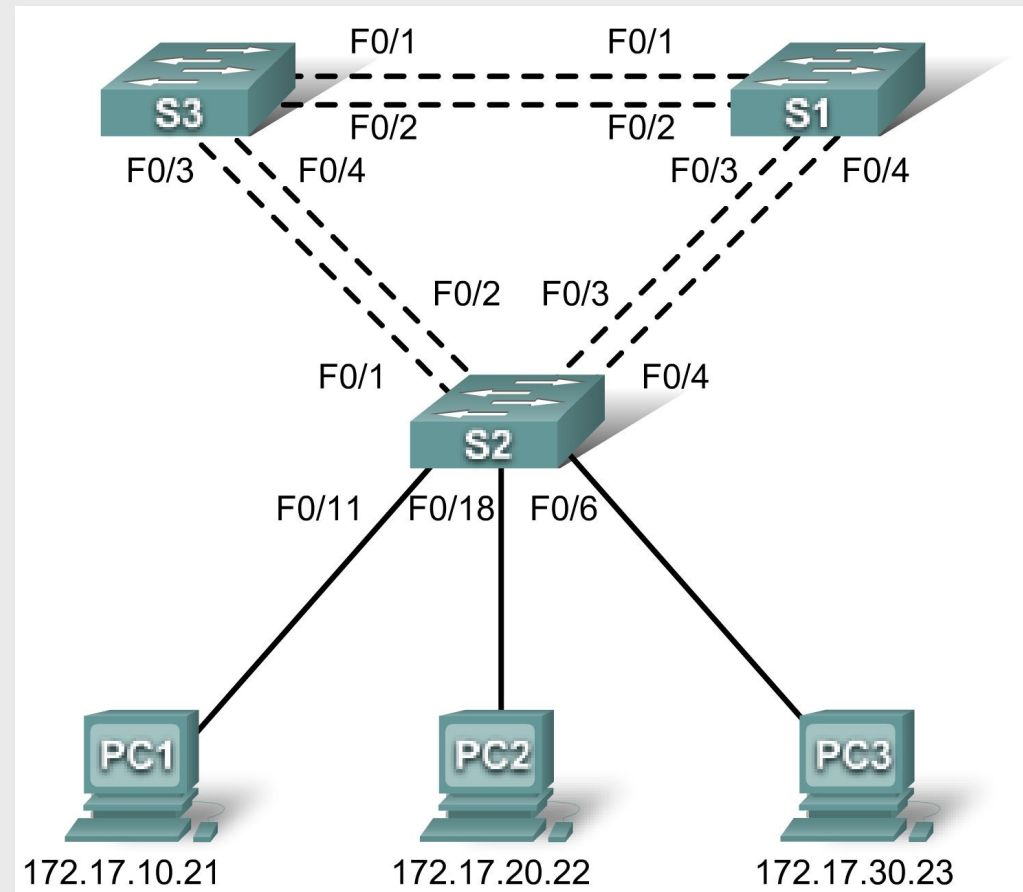
| Feature                                                             | Default Setting                                                                                                           |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Enable state                                                        | Enabled on VLAN 1                                                                                                         |
| Spanning-tree mode                                                  | PVST+ (Rapid PVST+ and MSTP are disabled.)                                                                                |
| Switch priority                                                     | 32768                                                                                                                     |
| Spanning-tree port priority (configurable on a per-interface basis) | 128                                                                                                                       |
| Spanning-tree port cost (configurable on a per-interface basis)     | 1000 Mb/s: 4, 100 Mb/s: 19, 10 Mb/s: 100                                                                                  |
| Spanning-tree VLAN port priority (configurable on a per-VLAN basis) | 128                                                                                                                       |
| Spanning-tree VLAN port cost (configurable on a per-VLAN basis)     | 1000 Mb/s: 4, 100 Mb/s: 19, 10 Mb/s: 100                                                                                  |
| Spanning-tree timers                                                | Hello time: 2 seconds<br>Forward-delay time: 15 seconds<br>Maximum-aging time: 20 seconds<br>Transmit hold count: 6 BPDUs |

# Lab activity

- Note: configure VLANs manually (instead of using VTP)



# Lab activity



# Lab activity

- Note: configure VLANs manually (instead of using VTP)

